

Modbus Communication Protocol: EL.T01 Modular Electronic Load

1.0 Introduction

The EL.T01 Modular Electronic Load system is designed as a master-slave architecture using the Modbus RTU protocol. The main controller module (EL.T01.MBRD) serves as the Modbus Master, communicating with multiple specialized slave modules over a shared RS-485 bus. Each slave module is assigned a unique Modbus slave address (1-247) via the Sequential_ID register.

This documentation provides a comprehensive register map for all modules, enabling scalable control, monitoring, and configuration. The system supports adjustable and stable electronic loads, precise measurements, battery management, and user interfaces.

Modbus Physical Layer Parameters:

- Protocol: Modbus RTU
- Baud Rate: 115200
- Data Bits: 8
- Parity: Odd
- Stop Bits: 1
- Flow Control: None
- Bus Topology: RS-485 half-duplex

2.0 General Principles

Addressing

Each slave module has its own independent register address space. To facilitate organization and future expansion, register addresses are divided into common and module-specific blocks. Common registers (applicable to all modules) use low address ranges (e.g., Coils 00001-00099, Input Registers 30001-30099). Module-specific registers start from higher blocks, allocated uniquely per module type for logical separation:

- EL.T01.PLAM: Coils 00100-00199, Discrete Inputs 10100-10199, Input Registers 30100-30199, Holding Registers 40100-40199
- EL.T01.PLSM: Coils 00200-00299, Discrete Inputs 10200-10299, Input Registers 30200-30299, Holding Registers 40200-40299
- EL.T01.MSCH: Coils 00300-00399, Discrete Inputs 10300-10399, Input Registers 30300-30399, Holding Registers 40300-40399
- EL.T01.KVMM: Coils 00400-00499, Discrete Inputs 10400-10499, Input Registers 30400-30499, Holding Registers 40400-40499
- EL.T01.CVCM: Coils 00500-00599, Discrete Inputs 10500-10599, Input Registers 30500-30599, Holding Registers 40500-40599
- EL.T01.UIM: Coils 00600-00699, Discrete Inputs 10600-10699, Input Registers 30600-30699, Holding Registers 40600-40699

- EL.T01.UPSM: Coils 00700-00799, Discrete Inputs 10700-10799, Input Registers 30700-30799, Holding Registers 40700-40799
- EL.T01.CHGM: Coils 00800-00899, Discrete Inputs 10800-10899, Input Registers 30800-30899, Holding Registers 40800-40899

This block allocation reserves ~100 addresses per category per module, allowing for future additions without overlap.

Data Types

Registers are 16-bit wide. Multi-word data types span consecutive registers:

- UINT32/INT32: 32-bit unsigned/signed integer (2 registers, big-endian: high word first).
 - FLOAT32: 32-bit IEEE 754 floating-point (2 registers, big-endian).
 - STRING: ASCII characters, 2 per register (high byte first), null-terminated and padded.
- Byte order (endianness): Big-endian (most significant byte first) for all multi-byte values.

Scaling

To represent physical quantities precisely without floating-point hardware, scaling factors are used:

- Currents: Stored as INT32/UINT32 in 0.001 A units (mA resolution).
- Voltages: Stored as INT32/UINT32 in 0.01 V units (10 mV resolution).
- Powers: Stored as UINT32 in 0.1 W units.
- Resistances: Stored as UINT32 in 0.01 Ω units.
- Temperatures: Stored as INT16 in 0.1 $^{\circ}\text{C}$ units (e.g., 25.5 $^{\circ}\text{C}$ = 255).
- Capacities/Percentages: Stored as UINT16 in % units (0-100).
- Times: Stored as UINT16 in minutes or seconds, as specified.
- Enums and bitmasks: UINT16, with defined values/flags.

Common Registers

The following registers are identical across all slave modules for consistency.

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).

Address Range	Register Name	R/W	Data Type & Scaling	Description
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A for details).
30003	Primary_Parameter	Read	INT32 (scaling varies)	Main measured value (scaling depends on module: e.g., current in 0.001A for loads/shunts, voltage in 0.01V for voltage modules) (spans 30003-30004).
30005	Max_Value	Read	UINT32 (scaling varies)	Maximum design value for primary parameter (same scaling as Primary_Parameter) (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.

Address Range	Register Name	R/W	Data Type & Scaling	Description
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.PLAM\0"), padded with nulls (20 chars max).
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.

3.0 Module Register Maps

3.1 EL.T01.PLAM - Adjustable Power Load

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0). Common to all modules.
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0). Common to all modules.
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0). Part of expanded status. Common.
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1). Part of expanded status. Common.
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1). Expanded for loads.

Address Range	Register Name	R/W	Data Type & Scaling	Description
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1). Added for safety.
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1). Assumes cooling fan; common for heat-generating modules.
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1). Added for reliability. Common.
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1). Added for persistence. Common.
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved. Common, but expanded per module.
30002	Error_Code	Read	UINT16	Error code (0=no error, 1=overheat, 2=overcurrent, etc.; define a table in docs). Common.
30003	Primary_Parameter	Read	INT32 (0.001 units)	Main measured value: Actual Current in 0.001A units (spans 30003-30004). Common, but for PLAM it's current.
30005	Max_Value	Read	UINT32 (0.001 units)	Maximum design value for primary parameter (current) in 0.001A units (spans 30005-30006). Common.
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number. Common.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number. Common.

Address Range	Register Name	R/W	Data Type & Scaling	Description
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23). Added identification. Common.
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00). Added identification. Common.
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.PLAM\0"), padded with nulls (20 chars max). Added identification. Common.
30101	Actual_Current	Read	INT32 (0.001A units)	Measured load current in 0.001A units (spans 30101-30102). PLAM-specific.
30103	Actual_Voltage	Read	INT32 (0.01V units)	Measured load voltage in 0.01V units (spans 30103-30104). Added as essential for loads.
30105	Actual_Power	Read	UINT32 (0.1W units)	Calculated power in 0.1W units (spans 30105-30106). PLAM-specific.
30107	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units. PLAM-specific, but common pattern.
30108	Fan_Speed	Read	UINT16 (RPM)	Fan speed in RPM (0 if no fan). Added for monitoring.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247). Common.
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults. Added for actions. Common.
40101	Load_Type	Read/Write	UINT16 (enum)	Load mode: 0=CC, 1=CV, 2=CR, 3=CP. PLAM-specific.
40102-40103	Target_Current	Read/Write	INT32 (0.001A units)	Target for CC mode in 0.001A units (spans 40102-40103). PLAM-specific.

Address Range	Register Name	R/W	Data Type & Scaling	Description
40104-40105	Target_Voltage	Read/Write	INT32 (0.01V units)	Target for CV mode in 0.01V units (spans 40104-40105). PLAM-specific.
40106-40107	Target_Resistance	Read/Write	UINT32 (0.01Ω units)	Target for CR mode in 0.01Ω units (spans 40106-40107). Added for CR mode.
40108-40109	Target_Power	Read/Write	UINT32 (0.1W units)	Target for CP mode in 0.1W units (spans 40108-40109). PLAM-specific.
40110-40111	Max_Current_Limit	Read/Write	UINT32 (0.001A units)	Overcurrent protection limit in 0.001A units (spans 40110-40111). Added configuration.
40112-40113	Max_Voltage_Limit	Read/Write	UINT32 (0.01V units)	Overvoltage protection limit in 0.01V units (spans 40112-40113). Added configuration.
40114	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units. Added configuration.
40115-40116	Calibration_Coeff_Current	Read/Write	FLOAT32	Calibration multiplier for current (IEEE 754 float, spans 40115-40116). Added for accuracy.
40117-40118	Calibration_Coeff_Voltage	Read/Write	FLOAT32	Calibration multiplier for voltage (spans 40117-40118). Added.
40119	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples). Added for noise reduction.

3.2 EL.T01.PLSM - Stable Power Load

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).

Address Range	Register Name	R/W	Data Type & Scaling	Description
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.001A units)	Main measured value: Actual Current in 0.001A units (spans 30003-30004).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30005	Max_Value	Read	UINT32 (0.001A units)	Maximum design value for primary parameter (current) in 0.001A units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.PLSM\0"), padded with nulls (20 chars max).
30201	Actual_Current	Read	INT32 (0.001A units)	Measured load current in 0.001A units (spans 30201-30202).
30203	Actual_Voltage	Read	INT32 (0.01V units)	Measured load voltage in 0.01V units (spans 30203-30204).
30205	Actual_Power	Read	UINT32 (0.1W units)	Calculated power in 0.1W units (spans 30205-30206).
30207	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
30208	Fan_Speed	Read	UINT16 (RPM)	Fan speed in RPM (0 if no fan).
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).

Address Range	Register Name	R/W	Data Type & Scaling	Description
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40201-40202	Target_Power	Read/Write	UINT32 (0.1W units)	Target power for stable load in 0.1W units (spans 40201-40202).
40203-40204	Max_Current_Limit	Read/Write	UINT32 (0.001A units)	Overcurrent protection limit in 0.001A units (spans 40203-40204).
40205-40206	Max_Voltage_Limit	Read/Write	UINT32 (0.01V units)	Overvoltage protection limit in 0.01V units (spans 40205-40206).
40207	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.
40208-40209	Calibration_Coeff_Power	Read/Write	FLOAT32	Calibration multiplier for power (spans 40208-40209).
40210	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).

3.3 EL.T01.MSCH - Shunt Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).

Address Range	Register Name	R/W	Data Type & Scaling	Description
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.001A units)	Main measured value: Current in 0.001A units (spans 30003-30004).
30005	Max_Value	Read	UINT32 (0.001A units)	Maximum design value for current in 0.001A units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.MSCH\0"), padded with nulls (20 chars max).
30301	Current	Read	INT32 (0.001A units)	Measured current in 0.001A units (spans 30301-30302).
30303	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40301-40302	Shunt_Resistance	Read/Write	UINT32 (0.0001Ω units)	Shunt resistance value in 0.0001Ω units (spans 40301-40302).
40303-40304	Calibration_Coeff_Current	Read/Write	FLOAT32	Calibration multiplier for current (spans 40303-40304).
40305	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).
40306	Max_Current_Limit	Read/Write	UINT32 (0.001A units)	Overcurrent detection threshold in 0.001A units (spans 40306-40307).
40308	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.

3.4 EL.T01.KVMM - Kelvin Voltage Measure Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.01V units)	Main measured value: Voltage in 0.01V units (spans 30003-30004).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30005	Max_Value	Read	UINT32 (0.01V units)	Maximum design value for voltage in 0.01V units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.KVMM\0"), padded with nulls (20 chars max).
30401	Voltage	Read	INT32 (0.01V units)	Measured Kelvin voltage in 0.01V units (spans 30401-30402).
30403	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40401	Voltage_Range	Read/Write	UINT16 (enum)	Measurement range: 0=low (e.g., 0-10V), 1=high (e.g., 0-100V).
40402-40403	Calibration_Coeff_Voltage	Read/Write	FLOAT32	Calibration multiplier for voltage (spans 40402-40403).
40404	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).

Address Range	Register Name	R/W	Data Type & Scaling	Description
40405-40406	Max_Voltage_Limit	Read/Write	UINT32 (0.01V units)	Overvoltage detection threshold in 0.01V units (spans 40405-40406).
40407	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.

3.5 EL.T01.CVCM - Cell Voltage Control Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).

Address Range	Register Name	R/W	Data Type & Scaling	Description
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.01V units)	Main measured value: Average cell voltage in 0.01V units (spans 30003-30004).
30005	Max_Value	Read	UINT32 (0.01V units)	Maximum design value for cell voltage in 0.01V units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.CVCM\0"), padded with nulls (20 chars max).
30501	Cell_Count	Read	UINT16	Number of cells monitored (e.g., 1-16).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30502	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
30503-30534	Cell_Voltages	Read	INT32[] (0.01V units)	Array of cell voltages, each INT32 spanning 2 registers (e.g., Cell 1: 30503-30504, Cell 2: 30505-30506, up to 16 cells). Unused cells read 0.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40501-40502	Calibration_Coeff_Voltage	Read/Write	FLOAT32	Global calibration multiplier for voltages (spans 40501-40502).
40503	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).
40504-40505	Max_Voltage_Limit_Per_Cell	Read/Write	UINT32 (0.01V units)	Overvoltage limit per cell in 0.01V units (spans 40504-40505).
40506	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.

3.6 EL.T01.UIM - User Interface Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).

Address Range	Register Name	R/W	Data Type & Scaling	Description
00601	Buzzer_Enable	Read/Write	Coil (BOOL)	Activate buzzer (1=on, 0=off).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
10601-10616	Button_Status	Read	Discrete Input[] (BOOL)	Status of up to 16 buttons/keypad inputs (1=pressed, 0=not; e.g., 10601=Button1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bits 7-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (N/A)	Reserved for future use (spans 30003-30004).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30005	Max_Value	Read	UINT32 (N/A)	Reserved for future use (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.UIM\0"), padded with nulls (20 chars max).
30601	Button_Bitmask	Read	UINT16 (bitmask)	Bitmask of all button statuses (Bit 0=Button1, etc.).
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40601	Backlight_Level	Read/Write	UINT16 (%)	Backlight brightness (0-100%).
40602	Buzzer_Duration	Read/Write	UINT16 (seconds)	Buzzer activation duration in seconds.
40603-40612	Display_Line_1	Read/Write	STRING (ASCII, 2 chars per reg)	Text for LCD line 1 (20 chars max, spans 40603-40612).
40613-40622	Display_Line_2	Read/Write	STRING (ASCII, 2 chars per reg)	Text for LCD line 2 (20 chars max, spans 40613-40622).

Address Range	Register Name	R/W	Data Type & Scaling	Description
40623-40632	Display_Line_3	Read/Write	STRING (ASCII, 2 chars per reg)	Text for LCD line 3 (20 chars max, spans 40623-40632).
40633-40642	Display_Line_4	Read/Write	STRING (ASCII, 2 chars per reg)	Text for LCD line 4 (20 chars max, spans 40633-40642).

3.7 EL.T01.UPSM - Uninterruptible Power Supply Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).

Address Range	Register Name	R/W	Data Type & Scaling	Description
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
10701	On_Battery	Read	Discrete Input (BOOL)	UPS is running on battery (1) or mains (0).
10702	Battery_Low	Read	Discrete Input (BOOL)	Battery capacity below threshold (1).
10703	Battery_Fault	Read	Discrete Input (BOOL)	Battery fault detected (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bit 7: On_Battery, Bit 8: Battery_Low, Bit 9: Battery_Fault, Bits 10-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.01V units)	Main measured value: Output Voltage in 0.01V units (spans 30003-30004).
30005	Max_Value	Read	UINT32 (0.01V units)	Maximum design value for output voltage in 0.01V units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.UPSM\0"), padded with nulls (20 chars max).
30701	Remaining_Capacity	Read	UINT16 (%)	Battery remaining capacity (0-100%).
30702	Output_Voltage	Read	INT32 (0.01V units)	Output voltage in 0.01V units (spans 30702-30703).
30704	Input_Voltage	Read	INT32 (0.01V units)	Input (mains) voltage in 0.01V units (spans 30704-30705).
30706	Load_Current	Read	INT32 (0.001A units)	Load current in 0.001A units (spans 30706-30707).
30708	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
30709	Runtime_Remaining	Read	UINT16 (minutes)	Estimated runtime on battery in minutes.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults.
40701	Low_Battery_Threshold	Read/Write	UINT16 (%)	Low battery alarm threshold (0-100%).
40702-40703	Calibration_Coeff_Voltage	Read/Write	FLOAT32	Calibration multiplier for voltages (spans 40702-40703).
40704-40705	Calibration_Coeff_Current	Read/Write	FLOAT32	Calibration multiplier for current (spans 40704-40705).

Address Range	Register Name	R/W	Data Type & Scaling	Description
40706	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).
40707	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.

3.8 EL.T01.CHGM - Battery Charger Module

Address Range	Register Name	R/W	Data Type & Scaling	Description
00001	Enable	Read/Write	Coil (BOOL)	Turn the module's main function on (1) or off (0).
00002	Reset	Write	Coil (BOOL)	Trigger a software reset (write 1 to activate; auto-resets to 0).
10001	Ready	Read	Discrete Input (BOOL)	Module is operational and ready (1) or not (0).
10002	Overheated	Read	Discrete Input (BOOL)	Temperature exceeds limit (1).
10003	Overcurrent_Trip	Read	Discrete Input (BOOL)	Overcurrent protection tripped (1).
10004	Overvoltage_Trip	Read	Discrete Input (BOOL)	Overvoltage protection tripped (1).
10005	Fan_Failure	Read	Discrete Input (BOOL)	Fan malfunction detected (1).
10006	Calibrated	Read	Discrete Input (BOOL)	Module has valid calibration (1).

Address Range	Register Name	R/W	Data Type & Scaling	Description
10007	Config_Saved	Read	Discrete Input (BOOL)	Configuration saved to non-volatile memory (1).
10801	Charging	Read	Discrete Input (BOOL)	Charging in progress (1).
10802	Charge_Complete	Read	Discrete Input (BOOL)	Charge cycle complete (1).
10803	Battery_Fault	Read	Discrete Input (BOOL)	Battery fault detected (1).
30001	Status_Bits	Read	UINT16 (bitmask)	Consolidated status bitmask: Bit 0: Ready, Bit 1: Overheated, Bit 2: Overcurrent_Trip, Bit 3: Overvoltage_Trip, Bit 4: Fan_Failure, Bit 5: Calibrated, Bit 6: Config_Saved, Bit 7: Charging, Bit 8: Charge_Complete, Bit 9: Battery_Fault, Bits 10-15: Reserved.
30002	Error_Code	Read	UINT16	Error code (0=no error; see Appendix A).
30003	Primary_Parameter	Read	INT32 (0.001A units)	Main measured value: Actual Charge Current in 0.001A units (spans 30003-30004).
30005	Max_Value	Read	UINT32 (0.001A units)	Maximum design value for charge current in 0.001A units (spans 30005-30006).
30007	Absolute_ID_High	Read	UINT16	High word of 32-bit factory-set serial number.
30008	Absolute_ID_Low	Read	UINT16	Low word of 32-bit factory-set serial number.
30009	Firmware_Version	Read	UINT16 (0.01 units)	Firmware version (e.g., 123 = v1.23).

Address Range	Register Name	R/W	Data Type & Scaling	Description
30010	Hardware_Version	Read	UINT16 (0.01 units)	Hardware version (e.g., 100 = v1.00).
30011-30020	Module_Name	Read	STRING (ASCII, 2 chars per reg)	Module name string (e.g., "EL.T01.CHGM\0"), padded with nulls (20 chars max).
30801	Actual_Charge_Current	Read	INT32 (0.001A units)	Measured charge current in 0.001A units (spans 30801-30802).
30803	Actual_Charge_Voltage	Read	INT32 (0.01V units)	Measured charge voltage in 0.01V units (spans 30803-30804).
30805	Temperature	Read	INT16 (0.1°C units)	Module temperature in 0.1°C units.
30806	Charge_Stage	Read	UINT16 (enum)	Current charge stage: 0=Idle, 1=Bulk, 2=Absorption, 3=Float, 4=Equalize.
30807	Time_In_Stage	Read	UINT16 (minutes)	Time elapsed in current stage in minutes.
40001	Sequential_ID	Read/Write	UINT16	Modbus slave address (1-247).
40002	Command	Write	UINT16	Command trigger: 0=noop, 1=reset, 2=clear_errors, 3=save_config, 4=start_calibration, 5=restore_defaults, 6=start_charge, 7=stop_charge.
40801	Charge_Profile_ID	Read/Write	UINT16 (enum)	Charge profile: 0=Li-ion, 1=NiMH, 2=Lead-Acid, 3=Custom.
40802-40803	Charge_Current	Read/Write	INT32 (0.001A units)	Target charge current in 0.001A units (spans 40802-40803).
40804-40805	Charge_Voltage	Read/Write	INT32 (0.01V units)	Target charge voltage in 0.01V units (spans 40804-40805).

Address Range	Register Name	R/W	Data Type & Scaling	Description
40806	Max_Charge_Time	Read/Write	UINT16 (minutes)	Maximum charge time limit in minutes.
40807-40808	Termination_Current	Read/Write	INT32 (0.001A units)	Termination current for absorption stage in 0.001A units (spans 40807-40808).
40809-40810	Calibration_Coeff_Current	Read/Write	FLOAT32	Calibration multiplier for current (spans 40809-40810).
40811-40812	Calibration_Coeff_Voltage	Read/Write	FLOAT32	Calibration multiplier for voltage (spans 40811-40812).
40813	Filter_Settings	Read/Write	UINT16 (samples)	Measurement averaging filter (e.g., 1=no filter, 16=average 16 samples).
40814	Max_Temperature_Limit	Read/Write	INT16 (0.1°C units)	Overtemperature limit in 0.1°C units.

4.0 Appendix A: Error Codes

Code	Description
0	No error
1	Over-temperature
2	Over-current
3	Over-voltage
4	Under-voltage
5	Fan failure
6	Calibration invalid
7	Configuration not saved
8	Communication timeout
9	Battery fault
10	Charge stage error
11	Invalid command
12-65535	Reserved for module-specific errors

5.0 Appendix B: Communication Examples

All examples use Modbus RTU framing: [Slave Address] [Function Code] [Data] [CRC16 (low byte, high byte)].

Register addresses in frames are 0-based (e.g., Input Register 30101 is address 0x0064 in hex, since 30101 - 30001 = 100, but actually Modbus input regs start from 0 as 30001).

Example 1: Reading Actual Current from PLAM (address 5)

- Function: 04 (Read Input Registers)
- Start Address: 30101 (0x0064 in frame, but wait: 30101 corresponds to internal addr 100, since 30000=0, but convention: 3xxxx means addr xxxx-1.
Standard: For Input Reg 3xxxx, frame addr = xxxx-1.
So for 30101, addr = 100 (0x0064).
Read 2 registers for INT32.
- Request: 05 04 00 64 00 02 [CRC]
- Response (example: Current = 5.000A = 5000 in 0.001A = 0x1388, big-endian 0x0000 0x1388): 05 04 04 00 00 13 88 [CRC]

Example 2: Setting a 10.5W Constant Power load on PLSM (address 6)

- Function: 10 (Write Multiple Holding Registers)
- Start Address: 40201 (internal addr 201-1=200 = 0x00C8)
- Write 2 registers for UINT32 Target_Power = 10.5W = 105 in 0.1W = 0x00000069
- Request: 06 10 00 C8 00 02 04 00 00 00 69 [CRC]
- Response: 06 10 00 C8 00 02 [CRC]

Example 3: Writing a charge profile to the EL.T01.CHGM module (address 7)

- Function: 06 (Write Single Holding Register)
- Register Address: 40801 (Charge_Profile_ID, internal addr 800 = 0x0320, since 40801 - 40001 = 800, 800 hex = 0x0320)
- Data: 0x0002 (enum value 2 for Lead-Acid profile)
- Request Frame (hex): 07 06 03 20 00 02 48 22 (Slave Addr: 07, Func: 06, Addr Hi: 03, Addr Lo: 20, Data Hi: 00, Data Lo: 02, CRC Lo: 48, CRC Hi: 22)
- Response Frame (hex): 07 06 03 20 00 02 48 22 (Echo of request, confirming write)

Example 4: Reading the operational status from the EL.T01.MSCH module (address 3)

- Function: 04 (Read Input Registers)
- Start Address: 30001 (Status_Bits and Error_Code, internal addr 0 = 0x0000)
- Number of Registers: 2
- Request Frame (hex): 03 04 00 00 00 02 70 29 (Slave Addr: 03, Func: 04, Start Addr Hi: 00, Start Addr Lo: 00, Count Hi: 00, Count Lo: 02, CRC Lo: 70, CRC Hi: 29)
- Response Frame (hex, example data: Status_Bits=0x0001, Error_Code=0x0000): 03 04 04 00 01 00 00 89 84 (Slave Addr: 03, Func: 04, Byte Count: 04, Data: 00 01 00 00, CRC Lo: 89, CRC Hi: 84)

6.0 Optimized Software Implementation for STM32L010K4T6 Modbus Slave

This section provides an optimized C code example for implementing a Modbus RTU slave on the STM32L010K4T4 microcontroller. The code uses direct register manipulation with CMSIS device headers for minimal overhead, an event-driven approach with interrupts, and a state machine for efficient parsing. Assumptions: System clock at 32 MHz (HSI + PLL), USART2 on PA2 (TX)/PA3 (RX) with AF4, TIM2 for timeout (334 μ s for 3.5 chars at 115200 baud). Global registers are static arrays in RAM, constants in Flash. CRC is computed with bitwise operations.

6.1 Optimized UART and Timer Configuration

```
#include "stm32l010xx.h" // CMSIS device header for register definitions

// Static globals for minimal stack use
static uint16_t holding_regs[100]; // Example holding registers (adjust size)
static uint16_t input_regs[100];   // Example input registers
static uint8_t modbus_buf[32];     // Small buffer for frames (max expected
size)
static uint8_t buf_idx = 0;        // Buffer index
static uint8_t state = 0;          // State machine: 0=idle, 1=recv, 2=process
static const uint8_t slave_addr = 1; // Const in Flash

// CRC16 table (const in Flash for memory efficiency)
static const uint16_t crc_table[16] = {
    0x0000, 0xCC01, 0xD801, 0x1400, 0xF001, 0x3C00, 0x2800, 0xE401,
    0xA001, 0x6C00, 0x7800, 0xB401, 0x5000, 0x9C01, 0x8801, 0x4400
};

// Optimized CRC16 computation (bitwise, no loop unrolling needed)
static uint16_t Modbus_CRC16(const uint8_t *data, uint8_t len) {
    uint16_t crc = 0xFFFF;
    while (len--) {
        uint8_t tmp = *data++ ^ crc;
        crc = (crc >> 8) ^ crc_table[tmp & 0x0F] ^ (crc_table[(tmp >> 4) & 0x0F] <<
4);
    }
    return crc;
}

void Init_USART2(void) {
    // Enable clocks (bitwise set)
    RCC->IOPENR |= RCC_IOPENR_GPIOAEN;
    RCC->APB1ENR |= RCC_APB1ENR_USART2EN;

    // GPIO PA2 (TX AF4), PA3 (RX AF4): alternate function mode
    GPIOA->MODER = (GPIOA->MODER & ~ (GPIO_MODER_MODE2_Msk |
GPIO_MODER_MODE3_Msk)) |
        (GPIO_MODER_MODE2_1 | GPIO_MODER_MODE3_1); // Mode 10b
    GPIOA->AFR[0] = (GPIOA->AFR[0] & ~ (GPIO_AFR_L_AFSEL2_Msk |
```

```

GPIO_AFRL_AFSEL3_Msk)) |
    (4UL << GPIO_AFRL_AFSEL2_Pos) | (4UL <<
GPIO_AFRL_AFSEL3_Pos);

// USART2 config: disable first
USART2->CR1 = 0;

// Set 8 data + parity (9 bits total), odd parity, enable RXNE interrupt
USART2->CR1 = USART_CR1_M0 | // M[1:0]=01 for 9 bits
    USART_CR1_PCE | USART_CR1_PS | // Parity enable, odd
    USART_CR1_RXNEIE; // RX interrupt enable

// 1 stop bit (default STOP=00)
USART2->CR2 = 0;

// Baud rate 115200 @ 32 MHz APB1: BRR = 32e6 / 115200 ≈ 278
USART2->BRR = 278;

// Enable TX, RX, USART
USART2->CR1 |= USART_CR1_TE | USART_CR1_RE | USART_CR1_UE;

// NVIC enable for USART2 IRQ
NVIC_SetPriority(USART2_IRQn, 0);
NVIC_EnableIRQ(USART2_IRQn);
}

void Init_TIM2(void) {
    // Enable clock
    RCC->APB1ENR |= RCC_APB1ENR_TIM2EN;

    // Prescaler for 1 MHz tick (32 MHz / 32 = 1 MHz, PSC=31)
    TIM2->PSC = 31;

    // Auto-reload for 334 us timeout (3.5 chars @ 115200)
    TIM2->ARR = 333; // Counts 0 to 333 = 334 ticks

    // Enable update interrupt
    TIM2->DIER = TIM_DIER_UIE;

    // NVIC for TIM2
    NVIC_SetPriority(TIM2_IRQn, 0);
    NVIC_EnableIRQ(TIM2_IRQn);
}

```

6.2 Optimized Modbus Protocol Handler

```

// State machine states
#define STATE_IDLE 0
#define STATE_RECV 1
#define STATE_PROC 2

```

```

// In UART ISR: receive byte, reset timer
void USART2_IRQHandler(void) {
    if (USART2->ISR & USART_ISR_RXNE) { // RX not empty
        uint8_t byte = USART2->RDR; // Read clears flag

        if (state == STATE_IDLE) {
            if (byte == slave_addr) { // Address match
                modbus_buf[0] = byte;
                buf_idx = 1;
                state = STATE_RECV;
            }
            // Else ignore
        } else if (state == STATE_RECV && buf_idx < sizeof(modbus_buf)) {
            modbus_buf[buf_idx++] = byte;
        }

        // Reset and start timer for frame timeout
        TIM2->CNT = 0;
        TIM2->SR = 0; // Clear UIF
        TIM2->CR1 = TIM_CR1_CEN; // Enable timer
    }

    // Handle TXE/TC if sending (in process)
    if (state == STATE_PROC && (USART2->ISR & USART_ISR_TXE)) {
        // Implement send logic if response ready (not shown, assume in
        Process_Modbus)
    }
}

// In TIM ISR: frame end, process
void TIM2_IRQHandler(void) {
    if (TIM2->SR & TIM_SR_UIF) {
        TIM2->SR = 0; // Clear flag
        TIM2->CR1 = 0; // Stop timer

        if (state == STATE_RECV && buf_idx >= 4) { // Min frame size
            state = STATE_PROC;
            Process_Modbus();
        }
        state = STATE_IDLE;
        buf_idx = 0;
    }
}

// Process frame and send response (minimal operations)
void Process_Modbus(void) {
    // Check CRC
    uint16_t rx_crc = (modbus_buf[buf_idx-1] << 8) | modbus_buf[buf_idx-2];
    if (Modbus_CRC16(modbus_buf, buf_idx-2) != rx_crc) return; // Invalid

    uint8_t func = modbus_buf[1];
    uint16_t addr = (modbus_buf[2] << 8) | modbus_buf[3];

```

```

uint8_t tx_buf[32]; // Small response buffer
uint8_t tx_idx = 2; // Start after addr+func
tx_buf[0] = slave_addr;
tx_buf[1] = func;

if (func == 0x03 || func == 0x04) { // Read holding/input
    uint16_t count = (modbus_buf[4] << 8) | modbus_buf[5];
    if (count > 0 && addr + count < 100) { // Range check
        tx_buf[2] = count * 2; // Byte count
        uint16_t *regs = (func == 0x03) ? holding_regs : input_regs;
        for (uint16_t i = 0; i < count; i++) {
            uint16_t val = regs[addr + i];
            tx_buf[tx_idx++] = val >> 8;
            tx_buf[tx_idx++] = val & 0xFF;
        }
    } else {
        tx_buf[1] |= 0x80; tx_buf[2] = 0x02; tx_idx = 3; // Exception
    }
} else if (func == 0x06) { // Write single
    uint16_t val = (modbus_buf[4] << 8) | modbus_buf[5];
    if (addr < 100) {
        holding_regs[addr] = val;
        tx_buf[2] = modbus_buf[2]; tx_buf[3] = modbus_buf[3];
        tx_buf[4] = modbus_buf[4]; tx_buf[5] = modbus_buf[5];
        tx_idx = 6;
    } else {
        tx_buf[1] |= 0x80; tx_buf[2] = 0x02; tx_idx = 3;
    }
} else if (func == 0x10) { // Write multiple
    uint16_t count = (modbus_buf[4] << 8) | modbus_buf[5];
    uint8_t byte_count = modbus_buf[6];
    if (byte_count == count * 2 && addr + count < 100) {
        for (uint16_t i = 0; i < count; i++) {
            holding_regs[addr + i] = (modbus_buf[7 + i*2] << 8) | modbus_buf[7 +
i*2 + 1];
        }
        tx_buf[2] = modbus_buf[2]; tx_buf[3] = modbus_buf[3];
        tx_buf[4] = modbus_buf[4]; tx_buf[5] = modbus_buf[5];
        tx_idx = 6;
    } else {
        tx_buf[1] |= 0x80; tx_buf[2] = 0x03; tx_idx = 3;
    }
} else {
    tx_buf[1] |= 0x80; tx_buf[2] = 0x01; tx_idx = 3; // Illegal func
}

// Append CRC and send byte by byte via TDR
uint16_t tx_crc = Modbus_CRC16(tx_buf, tx_idx);
tx_buf[tx_idx++] = tx_crc & 0xFF;
tx_buf[tx_idx++] = tx_crc >> 8;

// Send: enable TX interrupt

```

```

    USART2->CR1 |= USART_CR1_TXEIE;
    // Actual send in USART ISR on TXE (not shown fully, implement queue or
    direct)
}

// Main loop: empty for low power, all in ISRs
int main(void) {
    // System init (clock to 32MHz, etc., not shown)
    Init_USART2();
    Init_TIM2();
    while (1) {
        // Low power mode or other tasks
    }
}

```